

Lab Assignment 6

CS 331: Machine Learning

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Feb 10, 2025. You may work in groups.

This lab will be about EM for Gaussian Mixture Model (with multivariate data) with K Gaussians where K is an input. Questions 1-5 are mandatory. For multivariate data, the means, covariance matrices and the responsibility parameters π_j , $j \in \{1, \dots, K\}$ are the parameters. You need to submit your code as well as all the plots on classroom. Please include relevant plot titles and axis titles to help us see what each plot refers to.

1. Write a function to compute the log likelihood of the observed data $\log \mathbb{P}(\mathbf{X}|\Theta)$.
2. Implement EM algorithm for learning the parameters of GMM. The complete algorithm is provided in Christopher Bishop, Chapter 9, Section 9.2.2. Assign clusters to each point $\mathbf{x}_i$ using the posterior γ_{ik} as follows,

$$y_{il} = 1 \text{ for } l \in \arg \max_{l \in \{1, \dots, K\}} \gamma_{il}$$

Terminate EM algorithm when the log likelihood values do not change significantly.

3. Apply EM algorithm on the following two datasets attached:
 - OldFaithful dataset
 - dataset.csv

Standardize the datasets first. Use $K = 2$ for both datasets. For dataset.csv use initial parameters as follows: $\mu_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$, $\mu_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\Sigma_1 = \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\pi_1 = \pi_2 = 0.5$. For OldFaithful dataset, you may fix the initial centers as you did in Lab 4: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Compute the log likelihood in each iteration of EM and generate plots of how the log likelihood values change with each iteration.

4. For each dataset provide two plots showing the following: (a) the initial assignment of points to clusters along with the means (b) final assignment of points with cluster means [similar to Figure 9.1 in Christopher Bishop, Pattern Recognition].
5. Run K-means algorithm from Lab 4 on both datasets using the same starting means given in Question (3) and compare the final cluster assignments with what you got from EM. For this question you only need to submit the final cluster assignment plots from K-means.

ADDITIONAL QUESTIONS, EXPLORE FOR FUN!
YOU ARE NOT REQUIRED TO SUBMIT.

6. With different random starting parameters observe performance of EM and K-Means. Do you get different output cluster assignments with different starting parameters? What about the final log-likelihood values (EM objective) and distance of all points to centers (K-means objective)?
7. Run K-means algorithm and fix the centers obtained after K-means as the initial values of μ_1 and μ_2 for EM. Also initialize $\Sigma_1 = \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\pi_1 = \pi_2 = 0.5$. Run EM algorithm, how many iterations does it take to converge? Compare against the earlier starting point in Qn 3.
8. Run EM and K-means on the non-standardized datasets. Does standardization make a difference for these datasets?